ODD Semester Examination 2021-22
(Open Elective Course)

Course Code: OEENG103-**S** **Paper ID:**

Fundamental of Statistics

Time: 3 Hours **Max. Marks:** 60

Note: Attempt **all** questions.

<table>
<tr><td>Q1.</td><td>Answer the following (limit your answer to 50 words):</td><td>Marks</td></tr>
<tr><td>a)</td><td>What do you understand by primary data?</td><td>2</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>What do you understand by secondary data?</td><td>2</td></tr>
<tr><td>b)</td><td>Find the mean value of the following data:
10, 3, 8, 6, 4, 2, 7, 8.</td><td>2</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>What do you understand by Standard Deviation?</td><td>2</td></tr>
<tr><td>c)</td><td>Where is rank correlation used?</td><td>2</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>What is the purpose of multiple correlation?</td><td>2</td></tr>
<tr><td>d)</td><td>What is probability?</td><td>2</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>Define independent events.</td><td>2</td></tr>
<tr><td>e)</td><td>Define Bernoulli.</td><td>2</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>What is exponential?</td><td>2</td></tr>
<tr><td>2.</td><td>What are the different types of graphical representation of data?</td><td>10</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>Define statistics and also explain its characteristics in detail.</td><td>10</td></tr>
<tr><td>3.</td><td>What do you understand by skewness and kurtosis? In a sample in which conditions Skewness is shown.</td><td>10</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>What do you understand by dispersion measures and explain three measures of dispersion?</td><td>10</td></tr>
<tr><td>4.</td><td>Calculate the coefficient of correlation by rank difference method between the valuse of x and y from the following table:
<table><tr><td>X</td><td>77</td><td>88</td><td>99</td><td>70</td><td>60</td><td>80</td><td>60</td><td>60</td></tr><tr><td>Y</td><td>125</td><td>137</td><td>156</td><td>112</td><td>107</td><td>136</td><td>123</td><td>108</td></tr></table></td><td>10</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
<tr><td></td><td>Define the coefficient of correlation and regression coefficient. Also differentiate it?</td><td>10</td></tr>
<tr><td>5.</td><td>Differentiate between Independent events and algebra events.</td><td>10</td></tr>
<tr><td></td><td>Or</td><td></td></tr>
</table>

	Write the types of probability in detail.	10
6.	What is probability distribution? Write the functions of probability distribution in detail.	10
	Or	
	Explain the random variable. How it is differ with continuous variable?	10

Even Semester Examination 2021-22
(Open Elective Course)

Course Code: OEENG103-S **Paper ID:** 01122176

Fundamentals of Statistics

Time: 3 Hours **Max. Marks:** 60

Note: 1. Attempt **all** questions.

Q1.	Answer the following (limit your answer to 50 words):	Marks
a)	What is cumulative frequency curve?	2
	Or	
	State the difference between classification and tabulation of data.	2
b)	Distinguish between simple and weighted averages.	2
	Or	
	State the properties of arithmetic- Mean.	2
c)	Distinguish between multiple correlation and partial correlation.	2
	Or	
	How correlation is related with regression?	2
d)	Define conditional probability.	2
	Or	
	What is Addition Theorem of Probability?	2
e)	Define Probability Density Function.	2
	Or	
	State the properties of a normal distribution.	2
2.	a) What is scatter diagram? Interpret various types of correlation coefficient using scatter diagram. b) Define “sample space” and “event”. Write the sample space of the random experiment of throwing two dice	5+5=10
	Or	
	a) What are partition values? How are they determined graphically? Explain. b) What is a smooth frequency curve?How do you find it? State with the help of an example.	5+5=10
3.	a) Define weighted Arithmetic mean, Find the weighted arithmetic mean of first ‘n’ natural numbers, with weights being corresponding numbers. b) Derive the formula for mode in case of a continuous frequency distribution.	5+5=10
	Or	
	a) Define mean deviation and prove that is least when it is measured from the median. b) Define Kurtosis and prove that moment co-efficient of Kurtosis is independent of change of origin and scale.	5+5=10

4.	a) The mean of 5 observations is 4.4 and the variance is 8.24. If three of the five observations are 1, 2 and 6, find the values of the other two. b) The mean and median of a moderately skewed distribution are 42.2 and 41.9 respectively. Find the mode of the distribution.	5+5=10
	Or	
	a) What is the Harmonic Mean of 8, 1 and 6 weighted 3, 2 and 5 respectively? b) The mean and standard deviation of 200 items are found to be 60 and 20 respectively. If at the time of calculations, two items were wrongly recorded as 3 and 67 instead of 13and 17, find the correct mean and standard deviation.	5+5=10
5.	a) Define mutually exclusive events. b) Two cards are drawn from a pack of playing cards randomly. Find the probability of getting: i) Two kings ii) One spade and one club iii) An ace and a king	5+5=10
	Or	
	a) State Bayes theorem. b) Three urns contain 2 white and 3 black balls. 3 White & 2 black balls and 4 white and I black ball respectively. Oneball is drawn from an urn chosen at random and it was found to be white. Find the probability that it was drawn from the first urn.	5+5=10
6.	a) Suppose X is a Continuous random variable with p.d.f: f(x)=c (1-x2) for 0 <x< 1. Determine c and find the distribution function. b) Find the first four moments about the mean of the distribution having the p.d.f f(x)=f(x)=a/2; 0<x<a.	5+5=10
	Or	
	a) Obtain the recurrence relation for central moment of a binomial distribution. b) If x is a Poisson variate and P(O)=P(1)=c, then show that c=1/e.	5+5=10

Course Code: OEENG103	Fundamental of Statistics	L-3 T-0 P-0 C-3
Course Outcomes:	**On completion of the course, the students will be:**	
CO1.	**Understand** the concept of statistics, types of various data and their diagrammatical representation.	
CO2.	**Explain** the concept of measures of central tendency & dispersions.	
CO3.	**Relate** the concept of correlation & regression to estimate the relationship between two variables.	
CO4.	**Interpret** the concept of probability, laws of probability and Bayes' theorem with its applications.	
CO5.	**Apply** the concept of probability mass function, probability density function, distribution function on various continuous distribution.	
Course Content:		
Unit-1:	Introduction to statistics: Meaning, Definition, Characteristics. Data: classification and tabulationof data, diagrammatical representation of data, histograms, frequency polygons, smooth frequency polygons, cumulative frequency curve.	**8 Hours**
Unit-2:	Measures of central Tendency: Mean, Median, Mode. Measures of Dispersion: Range, Mean deviation, Standard deviation and Variance. Measures of Skewness and Kurtosis.	**8 Hours**
Unit-3:	Correlation: Introduction of correlation coefficient, Types of Correlation, Rank Correlation. Regression Analysis: Regression equations, Linear Regression, Relation between correlation coefficient and regression coefficients.	**8 Hours**
Unit-4:	Probability: Sample space, events and algebra of events, independent events. Kinds of Probability: classical, statistical, and axiomatic, Conditional Probability, laws of addition and multiplication and Bayes' Theorem.	**8 Hours**
Unit-5:	Probability distribution: random variable, discrete and continuous random variables, Probability mass function, Probability density function, Distribution functions. Bernoulli, Binomial, Poisson, Exponential, Uniform and Normal distribution.	**8 Hours**
Text Books:	1. Gupta, S. C. and Kapoor, V.K.: Fundamentals of Applied Statistics, Sultan Chand & Sons.	
Reference Books:	1. Goon A.M., Gupta M.K. and Dasgupta B.: Fundamentals of Statistics, The World Press, Kolkata. 2. Mood, A.M. Graybill, F.A. and Boes, D.C.: Introduction to the Theory of Statistics, Tata McGraw-Hill Pub. Co. Ltd. *** Latest editions of all the suggested books are recommended.**	